جامعة التقنية والعلوم التطبيقية
University of Technology and Applied Sciences
نزوى
Nizwa

Information Technology Department

Midterm Examination
Fall, AY 2022/2023
CSCM1101 – Fundamentals of Computing and Information System

Date: Tue, Nov 1, 2022 **Time:** 8:30 AM – 9:30 AM

Max Marks	20 Marks	Time Allocation	1 Hour

Student ID		Student Name	
Section		Level	Diploma Year I

Instructions to the Students

1. Answer the questions in the space provided by using **blue or black pen** only.
2. Use of mobile phones and any electronic gadgets inside the examination hall are not allowed.
3. ***Copying, Cheating, and any kind of malpractices on the examination are strictly prohibited.*** Action will be taken as per the "Article 80, Bylaws of Colleges of Technology".
4. Students should bring their own pencils, pens, erasers, rulers, calculators, and any other tools required for the exam.
5. **Use of standard calculators is not allowed for this exam.**
6. The exam has a total of **5 non-optional questions**.

Section	Allotted Marks	Obtained Marks
Section A	**8**	
Section B	**6**	
Section C	**6**	
Total Marks	**20**	

Marked By	Checked/Re-marked by

Total for Section A
/8

SECTION A. Multiple Choice Questions. (8 x 1 = 8 marks)
Direction: Read the statement in each question and choose the correct answer. Write the letter of your answer on the spaces provided below.

_____ 1. Windows 10 OS is an example of :
   a. System Software
   b. Application Software
   c. Knowledge Software
   d. Open Source Software

_____ 2. XYZ Company manufactures individual car parts such as pumps and engine. They sell it to an automotive company that builds the entire car from the parts.

Which category of e-commerce does the example above belongs?
   a. B2C
   b. B2B
   c. C2C
   d. C2B

_____ 3. In which component of Information System will hubs, switches and routers belong?
   a. People Resources
   b. Hardware Resources
   c. Software Resources
   d. Data Resources
   e. Network Resources

_____ 4. Which is an example of a general-purpose software?
   a. Anti-virus
   b. Mac OS
   c. Spreadsheet
   d. Student Registration System

_____ 5. It is a system used via the Internet or extranet for computer-to-computer exchange of e-commerce documents with business customers and suppliers.
   a. E-Commerce
   b. Electronic Data Interchange (EDI)
   c. Information System
   d. Online Payment

_____ 6. What business value of the telecommunication networks give information from any locations immediately after the information has been requested?
   a. Overcoming structural barriers
   b. Overcoming geographical barriers
   c. Overcoming time barriers
   d. Overcoming cost barriers

_____ 7. Which priority activity of the value chain model is responsible for receiving raw materials, inspection and storage?

a. Inbound logistics
b. Operations
c. Outbound logistics
d. Marketing and sales
e. Service

_____ 8. A Porter's Five Forces Model that is high when buyers have few choices from whom to buy and low when their choices are many.

a. Buyer Power
b. Supplier Power
c. Threat of Substitute Products or Service
d. Threat of New Entrants

Total for Section B
/6

SECTION B. Short Answer Questions. (4+2 = 6 marks)
Direction: Read each question and write your answers briefly in 1-3 sentences.

9. a. Why is a non-profit organization different from a profit organization? For profit organization, provide an example of a company that offers product and another example of a company that offers services. Provide two examples for non-profit organization. (3)

b. What are the primary production factors provided by the environment for an organization? (1)

10. a. Give an example of data and information. (1)

b. From the sample you have given in Q10a, describe how information can be applied to become knowledge. (1)

Total for Section C
/6

SECTION C. Case Scenario Questions. (2 x 3 = 6 marks)
Direction: Read the case scenario below and answer the questions based on each case scenario.

11. Serendipity Coffee Scenario

Serendipity Coffee, which is established in January 2000, is a coffee shop that offers coffee products and pastries to its customers in Nizwa. The shop has made improvements in their ordering and sales system by using a Point-of-Sale system used by each wait staff in a tablet, which connects to a server through hubs, routers and fiber optics. It is the first coffee shop that used a POS, which aims at improving customer experience by providing faster and efficient transactions. The shop also offers low-cost meals by buying raw materials directly from suppliers and by employing non-professional workers that they train after hiring. In September 2010, Serendipity Coffee diversified the products by launching the sale of coffee mugs, coffee beans, portable coffee makers and French press in their shop. In August 2022, Serendipity Coffee launched its website to allow consumers to order online and allow communication between the shop and the consumers.

A. Identify two competitive strategies mentioned in the scenario and explain how these strategies can help Serendipity Coffee to have a competitive advantage. (1 + 1)

B. List any two business value of telecommunication network found in the scenario. (.5 + .5)

12. Al Dhakliya Bank

Al Dhakliya Bank is one of the leading financial services providers in Oman providing corporate banking and retail banking. To open a savings account, the customer needs to submit the resident id and personal details. The details will be verified to check the correctness of data and avoid duplication. After verification, the customer will receive the debit card, account number and initial password. To use the debit card for shopping, the store attendant will insert the card in the card reader slot and enter the bill and PIN. The PIN will be validated. If the PIN is valid the account will be updated and the SMS will be received by the customer from the bank. In every transaction, the account of the customer is updated to ensure that the information is trustworthy and can be depended on.

A. Identify one characteristic of valuable information found in the scenario and briefly explain why do you consider it as a valuable information in relation to Al Dhakliya Bank. (1.5)

B. Identify the Input, Process and Output in opening a savings account. (1.5)